# VESPER

A PERSISTENT, DISTRIBUTED OPERATING PRESENCE

*The model may change. The record of what matters must endure.*

A personal assistant being engineered around durable state, evidence, bounded agency and continuity across replaceable models and devices. This plate maps the current source; deployment and hardware validation remain separate claims.

● CURRENT ● EXPERIMENTAL ● PLANNED

CURRENT = implemented source. EXPERIMENTAL = measurement in progress. PLANNED = specified or proposed, not established in operation.

## I / THE LIVE TURN

Solid arrows: calls / results  Orbit lines: visual metaphor

**OWNER INPUT / INTAKE**

typed text + final, confidence-gated speech

no action proposed: answer()

REQUEST

### 01 Choose the model

Difficulty chooses a profile; interactive turns prefer measured latency.

`chooser.py / router.py`

### 02 Assemble the situation

Persona, recent dialogue, live mode and selected facts beside the question.

`blocks.py / context.py`

### 03 Cross the gateway

Privacy, credential redaction and cost rules precede provider attempts.

`gateway.py / routing/supply.py`

### 04 Answer and retain

Reply to the owner; capture utterances asynchronously and record misses.

`answer() / updater.py`



VESPER

IDENTITY IS THE SYSTEM, NOT THE MODEL

Orchestrated by vesper_runtime.py

### 01 Propose one next step

A model names a known capability and arguments; the proposal grants no authority.

`intent.py / actor.py`

### 02 Resolve permission

Evaluate scoped grants, argument-sensitive levels and ceilings; ask when required.

`authorize.py / confirm.py`

### 03 Execute with a record

Re-check authority; checkpoint file changes and record external effects before acting.

`executor.py / checkpoint.py`

### 04 Observe and verify

Feed observations to the next step. Check effects where an independent verifier exists.

`actor.py / capabilities.py`

ACT LOOP: 6 STEPS MAX / 3 CONSECUTIVE REFUSALS STOP THE TURN

**TRANSCRIPT + SPEECH + EXPLANATION SURFACES**

*No generic shell-effect verifier: an unchecked result is reported as such.*

## II / WHAT MAKES AN ASSISTANT

*A composition of mechanisms; no claim of consciousness.*

**MODEL**

A replaceable inference service. It generates answers or proposes steps; its output is not permission.

**MEMORY**

Stored facts and their provenance. The log preserves history; the graph makes it selectable.

**CONTEXT**

The bounded material supplied now: rules, evidence, recent dialogue and the current request.

**TOOLS**

Concrete adapters that read or change the world. Arguments determine their required authority.

**AGENT**

A bounded task process with a purpose and tool scope. Advice returns with external-content taint.

**INTERFACE**

The native island and home, speech and typed scenes. They expose state and receive the owner.

**ORCHESTRATOR**

The runtime joins intake, models, memory and actions, while a separate loop evaluates world state.

### 01 Memory with a source

`memory.py / blocks.py / knowledge_map.py`

**L3** is the append-only event log. A fact is written there first; its returned sequence becomes the provenance pointer in **L2**, a SQLite graph of entities, facts and edges. **L1** is the context rendered for a turn.

**L3** EVENT LOG → **L2** FACT GRAPH → **L1** TURN CONTEXT

Selection uses **FTS5 + a graph walk bounded at depth 3**, then authority and recency. Block A carries a stable persona and index; block B places up to five relevant facts next to the question. Budget cuts and unresolved source pointers are reported.

The knowledge map navigates to hash-checked source spans; a map summary is not evidence. Compaction reprojects from L3. Erasing L2 does not destroy L3 history. No embedding store is used.

`RANK: rule > source record > observation > user > derived`

### 02 Authority before effect

`authorize.py / taint.py / permission_mode.py`

A capability name is insufficient: the same command tool can inspect a directory or destroy it. VESPER resolves authority from the actual arguments and the source of the instruction.

$$E = \min(G, D, C, M, A)$$

**E**: effective level. **G**: grant maximum. **D**: declared argument level. **C**: context-trust ceiling. **M**: owner-mode ceiling. **A**: authorization-strength ceiling. Levels are ordered L0 through L5.

This is an upper bound, not the whole permit: expiry, target scope, lease epoch, required level and further boundary checks still apply. Retrieved content stays tainted through summaries and caches. An advisor may add STOP; it cannot manufacture consent.

`SELF-MOD: propose and test; runtime self-merge is refused`

### 03 State can initiate work

`sensors.py / world.py / autonomy.py`

Host probes and agenda bridges supply observations. Facts expire to **UNKNOWN**; stale values do not masquerade as current ones. The deterministic trigger evaluator cannot call a model. Models can help carry out a decision after it fires.

`OBSERVE → WORLD STATE → PREDICATE → HANDLER`

Live wiring includes fact expiry, memory drift, contradiction handling, map staleness and due agenda items. Interruption budgets decide when to surface non-alarm work. A fresh external enable token gates autonomous actuation.

Events use **HLC = (physical_ms, logical, node_id)**, ordered lexicographically: physical time, then a logical counter, then a node tie-breaker. Wall-clock arrival order cannot supersede a fact. Consumers must declare idempotency keys.

`RECOVERY: fenced leases / durable tasks / checkpoints`

### 04 Voice, sight & touch

`Swift bridges / speech.py / transcribe.py`

**CURRENT** Typed input, on-device English recognition, a local wake-word listener and Fn activation. Darija transcription uses permitted remote suppliers; speech uses provider routing or local synthesis, with degradations reported.

Application and window adapters read, write, move and inspect. Accessibility actions re-resolve an element and check its role and label before touching it. UI-tree access is limited to the showing Desktop. Camera identity remains **PLANNED**.

### 05 Runtime & instruments

`build.sh / store.py / provider_supply.toml`

**CURRENT** Python core; SQLite persistence; Swift/AppKit host; Metal face and ambience beneath WebKit HTML/CSS/JS. The app launches its bundled runtime. Five physical stores separate log, audit, tombstones, state and workspace.

Supply contracts name Cloudflare, Groq, Gemini, Ollama and Mistral routes, plus purpose-specific voice and agent suppliers. These are configured candidates, not guaranteed availability. A bounded stdio MCP client is built; live server wiring is not established here.

### 06 A fabric, still becoming

`placement.py / fabric.py / SPIKE-001`

**CURRENT** Placement and survival decisions use measured capability profiles and trust constraints. Network trust can only weaken node trust. The dedicated segment remains **unattested**; deployment is not a demonstrated roaming fabric.

**EXPERIMENTAL** Local inference is profiled with llama.cpp. The recorded node-1 Qwen3-8B Q4_K_M run sustained **45.8 tok/s** (b10405-e79e4bf66, Vulkan; median of 3 x 10-minute runs, 2026-08-17). This is throughput, not conversational latency.

## III / BUILD RECORD & OPEN FRONTIER

**24 AUG 2026** — First routed model call

**30 AUG 2026** — L2 memory committed

**31 AUG 2026** — Voice + act path joined

**01–11 SEP 2026** — Multi-step agency and host control

**PLANNED / OPEN** Camera identity; cross-site connectivity; remote wake and service dormancy; multi-model residency; system widgets. SPIKE-002 still owes the full voice-latency chain. SPIKE-013 is measuring home-composite cost.

Evidence discipline: 104 / 122 check IDs implemented; 18 PENDING. Registry inspected 2026-09-11, not a fresh full gate run.

*To remain useful, remember the source. To act, return to the world.*